FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – IV

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 264**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into Three sections: **Section-A Section-B & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 10, 21 – 30, 41 – 50)** contains 30 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–B (11 – 12, 31 – 32, 51 – 52)** contains 6 Match the following Type questions. Each question having 4 statements in Column I & 5 statements in Column II with any given statement in Column I having correct matching with 1 or more statement (s) given in Column II. Each statement carries **+2 marks** for correct answer and **–1 mark** for wrong answer.

3. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

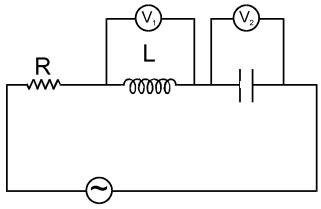
Physics

PART – I

SECTION – A
One OR More Than One Choice Type

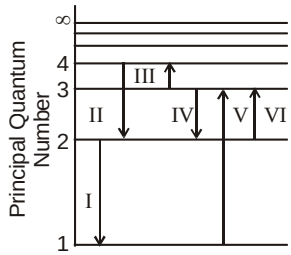
This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. In the figure shows $R = 100\ \Omega$, $L = \frac{2}{\pi}$ H and $C = \frac{8}{\pi}\ \mu F$ are connected in series with an a.c. source of 200 volt and frequency f. V_1 and V_2 are two hot-wire voltmeters. If the reading of V_1 and V_2 are same, then



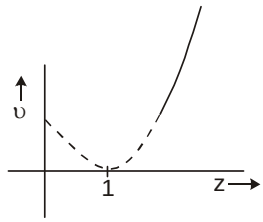
(A) f = 125 Hz
(B) $f = 250\pi$ Hz
(C) Current through R is 2 A
(D) $V_1 = V_2 = 1000$ volt

2. The figure shows an energy level diagram for the hydrogen atom. Several transitions are marked as I, II, III, ……. The diagram is only indicative and not to scale.



(A) The transition in which a Balmer series photon absorbed is VI
(B) The wavelength of the radiation involved in transition II is 486 nm
(C) IV transition will occur when a hydrogen atom is irradiated with radiation of wavelength 103 nm
(D) IV transition will emit the longest wavelength line in the visible portion of the hydrogen spectrum

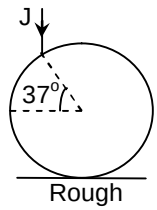
3. Figure shows the variation of frequency of a characteristic X-ray and atomic number.



(A) The characteristic X-ray is K_α
(B) The characteristic X-ray is K_β
(C) The energy of photon emitted when this X-ray is emitted by a metal having z = 101 is 204 keV
(D) The energy of photon emitted when this X-ray is emitted by a metal having z = 101 is 102 keV

Space for Rough work

4. A sharp vertical impulse J is given to a uniform solid sphere of mass m and radius R placed on a rough horizontal surface. Choose the correct options.



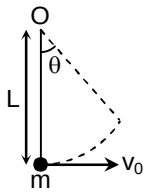
(A) Impulse imparted to the sphere by the surface perpendicular to the surface will also have magnitude J

(B) Minimum value of coefficient of friction between sphere and surface to avoid slipping is $\mu = \frac{4}{7}$

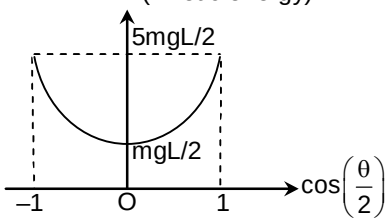
(C) If $\mu = \frac{11}{20}$, sphere rolls with backward slipping

(D) If $\mu = \frac{13}{20}$, sphere rolls with forward slipping

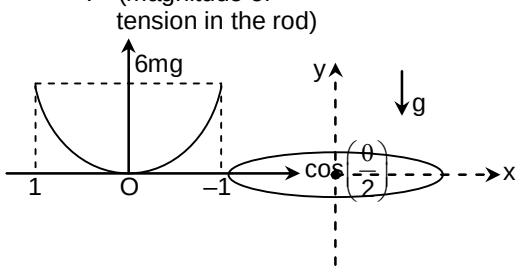
5. A massless rod hinged at one end free to rotate in vertical plane has a bob of mass m attached to it. Bob is given a velocity v_0 at the bottom most point. Choose the correct curves from the below options. $\left(v_0 = \sqrt{5gL}\right)$



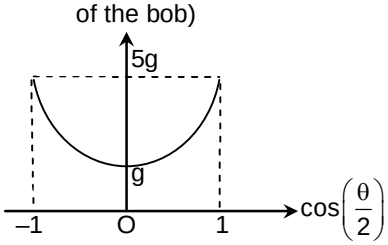
(A)



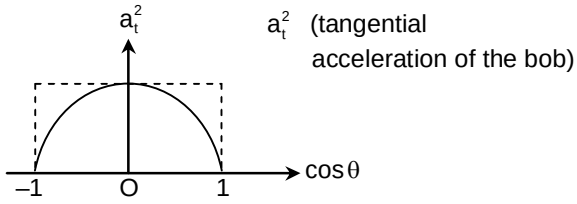
(B)



(C)



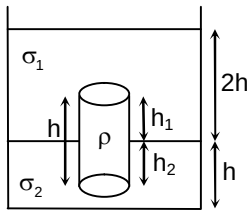
(D)



Space for Rough work

6. A uniformly charged ring having total charge Q and of radius R is fixed in such a way so that plane of the ring is horizontal. A charged particle of charge q and mass m is in equilibrium at any point lying on y-axis under the combined forces of electric field and earth's gravitational field. Choose the correct statements.
   (A) If $qQ > 0$, particle must be on +y-axis
   (B) If $qQ < 0$, particle must be on –y-axis
   (C) If the particle is in stable equilibrium magnitude of y-coordinate of particle must be greater than $\frac{R}{\sqrt{2}}$
   (D) If the particle is at y distance from centre of the ring. Specific charge of the particle is $\frac{4\pi\varepsilon_0\left(y^2+x^2\right)^{\frac{3}{2}}g}{Qy}$

7. A cylinder of uniform mass density ρ is in equilibrium under the pressure forces acting due to two ideal liquids of density σ_1 and σ_2. Choose the correct alternative.



   (A) This is only possible if $\sigma_1 > \rho > \sigma_2$
   (B) $\frac{h_2}{h_1} = \left|\frac{\sigma_1-\rho}{\rho-\sigma_2}\right|$
   (C) If the cylinder is pressed slightly towards the bottom surface, it performs periodic motion
   (D) If the cylinder is pressed slightly towards the bottom surface, it reaches at the bottom surface with velocity $\sqrt{\left|\frac{\sigma_1-\sigma_2}{\rho}\right|\frac{gh_1^2}{h}}$

8. Decay constant of a radioactive sample is 4.9×10^{-18} s^{-1}. Choose the correct statements regarding this sample.
   (A) Average life is 6.49×10^9 years
   (B) Half life is 4.5×10^9 years
   (C) Activity of the sample decreases by 4 in 9×10^9 years
   (D) 87.5% of the initial number of nuclei is decayed in time 13.5×10^9 years

Space for Rough work

9. A drum of mass m_1 and radius r_1 rotates freely with initial angular velocity ω_0. A second drum with mass m_2 and radius r_2 $(r_2 > r_1)$ is mounted on same axle and is at rest although it is free to rotate. A thin layer of sand with mass m is distributed on inner surface of smaller drum. At t = 0, small perforations in the inner drum are opened. The sand starts to fly out at a constant rate λ kg/sec and sticks to the outer drum. Ignore the transit time of the sand. Choose the correct alternatives.

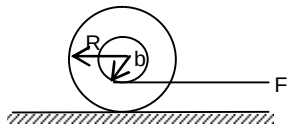
(A) Angular speed of outer drum at time t is $\dfrac{\lambda t\omega_0}{m_2+\lambda t}\left(\dfrac{r_1}{r_2}\right)^2$

(B) Difference in final angular speeds of two drums is 0.

(C) Difference in final angular speeds of two drums is $\left(\dfrac{m\left(r_2^2-r_1^2\right)+m_2r_2^2}{\left(m+m_2\right)r_2^{\,2}}\right)\omega_0$

(D) Angular speed of inner drum remains constant.

10. A Yo-Yo of mass m has an axle of radius b and a spool of radius R. The Yo-Yo is placed upright on a table and string is pulled with a horizontal force $\vec{F}$ to the right as shown in figure.



The static friction coefficient between Yo-Yo and table is μ_s. Now there are two cases.
case (a): The string is pulled very gently?
case (b): The string is jerked hard?
The Yo – Yo will rotate:
(A) Clockwise in case (a),
(B) Anticlockwise in case (a)
(C) Clockwise in case (b)
(D) Anticlockwise in case (b)

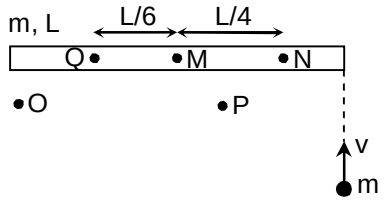
Space for Rough work

SECTION – B

(Matching Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct.

11. A uniform rod of length L lies on smooth horizontal table. A particle of same mass moving on table strikes the rod perpendicularly as shown in figure. Q, M, N are points on the rod and O & P are fixed at ground.



After collision if particle comes in rest,

Column I		**Column II**	
(A)	Angular momentum of system will be conserved about points	(p)	Q
(B)	Linear momentum conserved from points	(q)	M
(C)	Speed of point Q just after collision is greater than speed of points	(r)	N
(D)	If coefficient of restitution of collision is $e = \frac{1}{2}$, then angular momentum is conserved about points	(s)	O
		(t)	P

Space for Rough work

12. Match Column I with Column II:

Column I		Column II	
(A)	A circular conducting loop rotating about its axis in a uniform constant magnetic field perpendicular to its plane	(p)	Net induced emf in the loop is non zero
(B)	A circular conducting loop moving along its plane in pure translation in a uniform constant magnetic field perpendicular to its plane	(q)	Net induced emf is zero but a small part of the loop may have some emf induced across it
(C)	A circular conducting loop placed in a magnetic field perpendicular to its plane which is decreasing with time	(r)	Induced emf in any small part of the loop is zero
(D)	A square conducting loop rotating about its side in a uniform constant magnetic field perpendicular to its axis. Consider a time when the plane of the loop is parallel to the magnetic field	(s)	Induced electric field outside the conductor is zero
		(t)	Current flows in the loop

SECTION – C

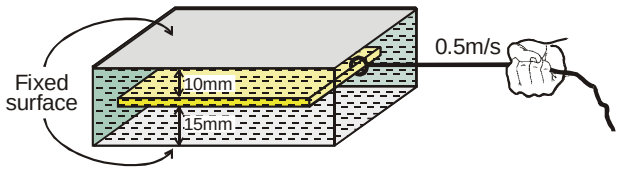
(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

13. 1 kg ice at –10°C is mixed with 0.1 kg of steam at 200°C. If final temperature of mixture at equilibrium is $T_{eq} = \frac{58x}{11}$, then fill the value of x.

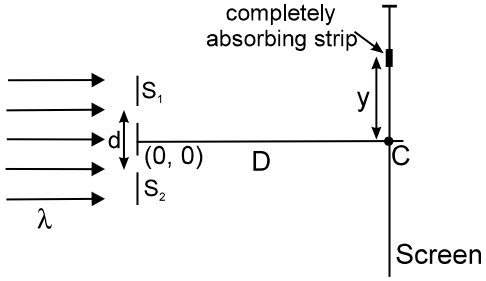
Latent heat of fusion of ice = 80 cal/gram, latent heat of vaporization of water = 540 cal/gram, specific heat capacity of ice $\cong$ specific heat of water = 0.5 cal/gram-K.

14. Glycerine is filled in 25 mm wide space between two large plane horizontal surfaces. Calculate the force required to drag a very thin plate 0.75 m^2 in area between the surface at a constant speed of 0.5 m/s if it is at a distance of 10 mm from one of the surfaces in horizontal position? Take coefficient of viscosity η = 0.5 Ns/m^2. Fill the value of X where X = force required to drag/6.25.

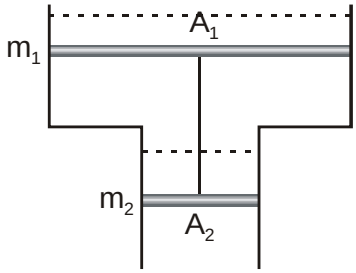


Space for Rough work

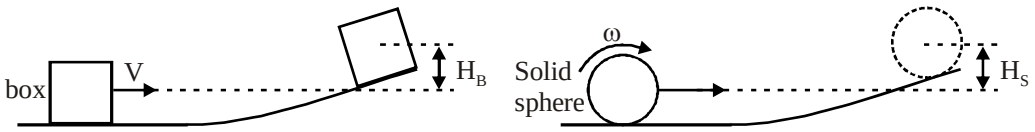
15. Figure shows two, identical narrow slits S_1 and S_2. A very small completely absorbing strip is placed at distance y from the point C. C is the point on the screen equidistant from S_1 and S_2. Assume $\lambda << d << D$, where λ, d and D have usual meaning. When S_2 is covered the force due to light acting on strip is f and when both slits are opened the force acting on strip is 2f. Find the minimum positive y (<< D) coordinate (in cm) of the strip if $\frac{\lambda D}{d}$ = 4 cm.



16. Consider a vertical tube open at both ends. The cube consists of two parts, each of different cross-sections and each part having a piston which can move smoothly in respective tubes. The two pistons are joined together by an inextensible wire. The combined mass of the two piston is 5 kg and area of cross-section of the upper piston is 10 cm^2 greater than that of the lower piston. Amount of gas enclosed by the pistons is one mole. When the gas is heated slowly, pistons move by 50 cm. Find rise in the temperature of the gas, in the form 25X/R K where R is universal gas constant. Use g = 10 m/s^2 and outside pressure = 10^5 N/m^2). Fill value of X in the answer sheet.

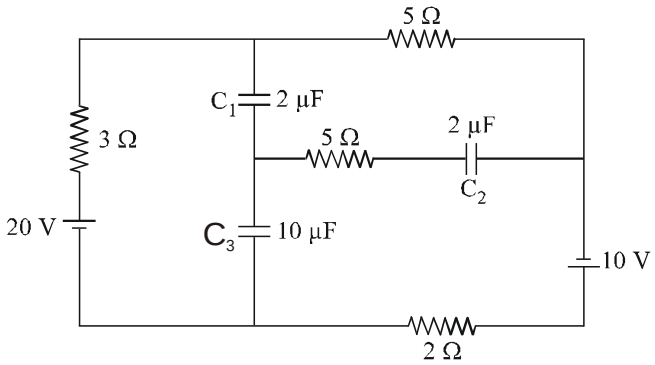


17. A box and a solid sphere of equal mass are moving with the same velocity across a horizontal floor. The sphere rolls without slipping and the box slides without friction. They encounter an upward slope in the floor and each move up the slope some distance before momentarily stopping and then moving down again. The maximum vertical heights reached by the box and the sphere are H_B and H_S respectively. What is the ratio $\frac{5H_S}{H_B}$?

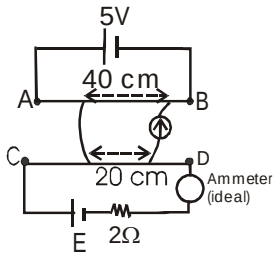


Space for Rough work

18. In the circuit shown in figure, if the charge on capacitor C_2 in steady state is $\frac{10x}{3}$ μC, then find value of x.



19. AB and CD are two uniform resistance wires of lengths 100 cm and 80 cm respectively. The connections are shown in the figure. The cell of emf 5 V is ideal while the other cell of emf E has internal resistance 2Ω. A length of 20 cm of wire CD is balanced by 40 cm of wire AB. If emf E in volt is 2x, then find the value of x, if the reading of the ideal ammeter is 2A. The other connecting wires have negligible resistance.



20. A radioactive nucleus can decay by three different processes. Half life for first process is 2 hours. Effective half life of the nucleus is $\frac{4}{3}$ hours. Find the half life for second process in hours.

Space for Rough work

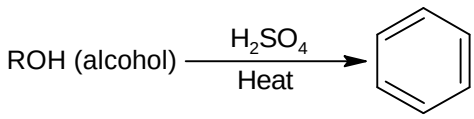
Chemistry

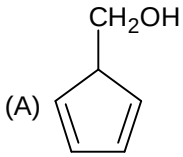
PART – II

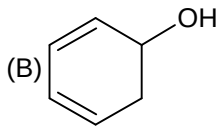
SECTION – A
One OR More Than One Choice Type

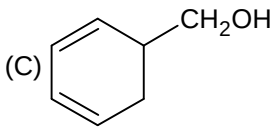
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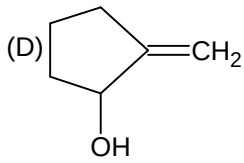
21. The possible structure of the alcohol is



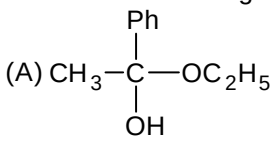


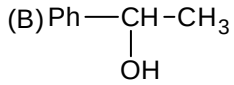


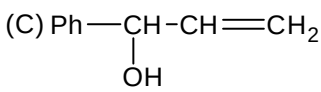


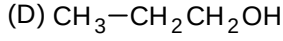


22. Which of the following alcohol gives the white turbidity immediately with HCl + $ZnCl_2$ (anhy.)?



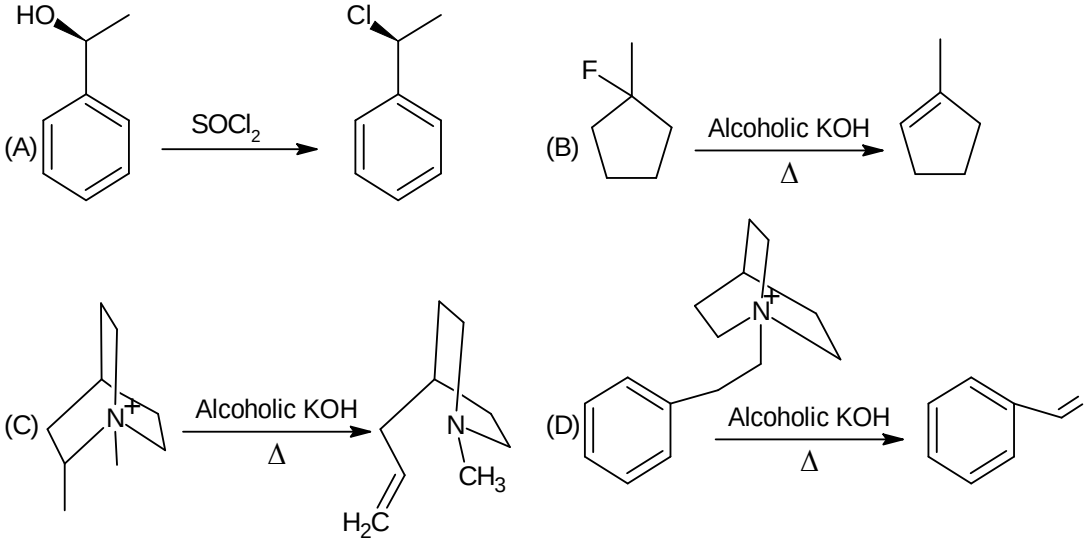




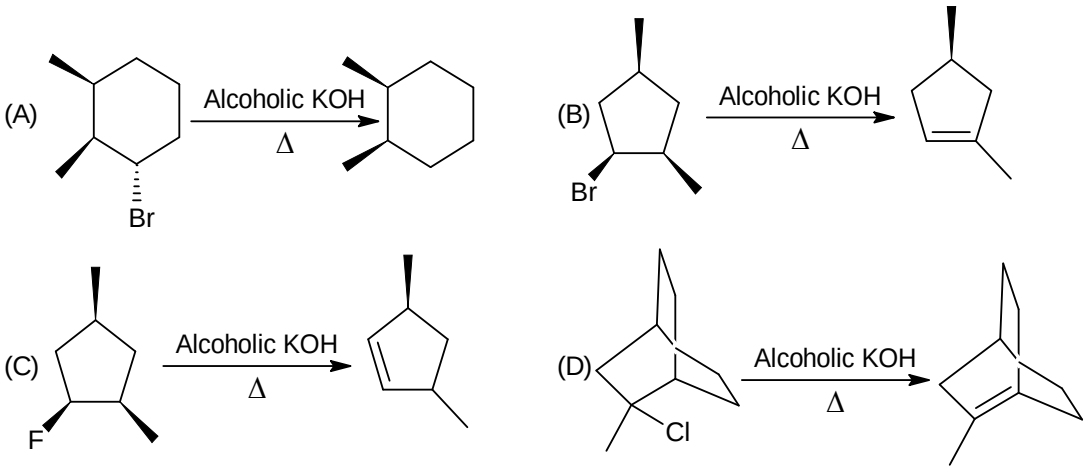


Space for Rough work

23. Identify reactions that correctly match with their products:

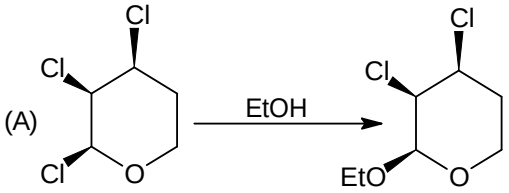


24. Identify reactions that correctly match with their products:

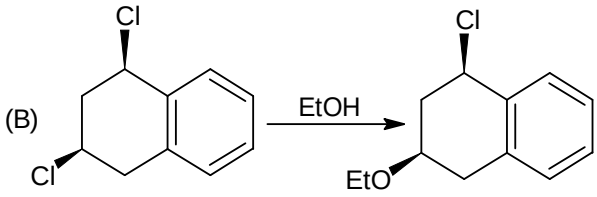


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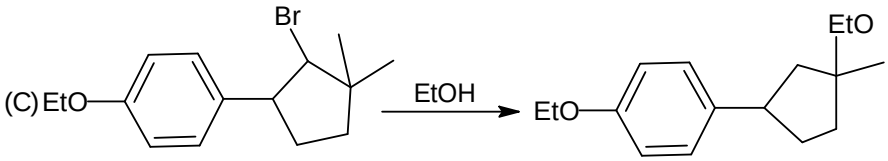
25. Identify reactions that correctly match with their products?

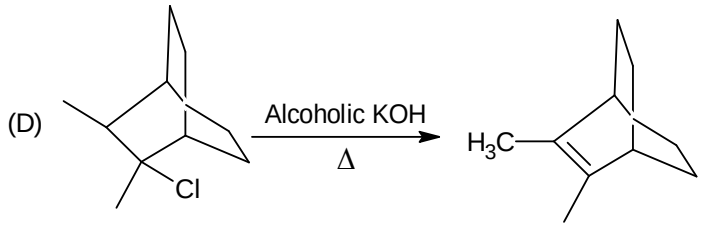


Monosubstitued product



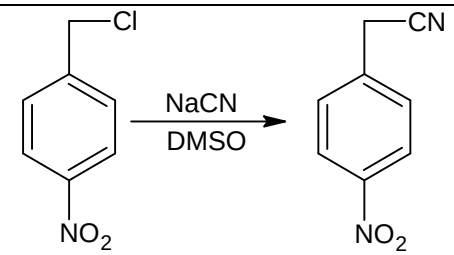
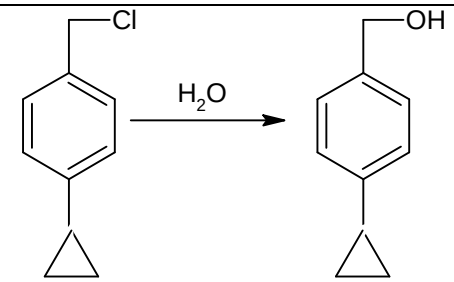
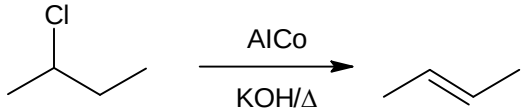
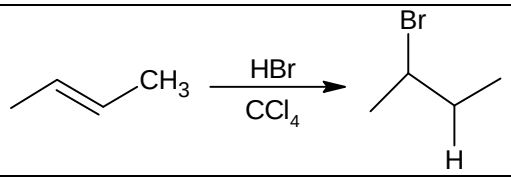
Monosubstitued product



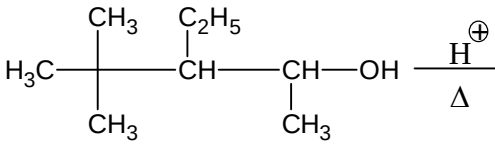


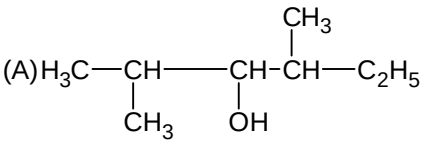
Space for Rough work

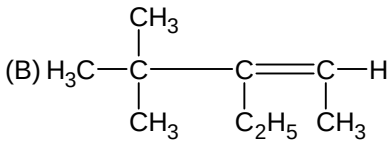
26. Which of the following is/are correctly matched?

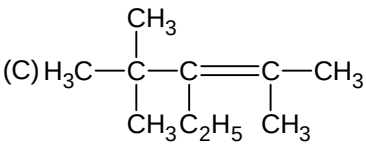
Column I		Column II	
(A)		(P)	Carbocation intermediate is formed
(B)		(Q)	S_N1
(C)		(R)	S_N2
(D)		(S)	Addition reaction

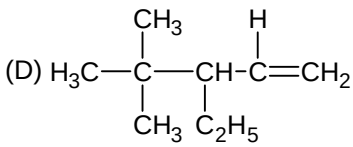
Space for Rough work

27.  Product mixture, the product's mixture contains

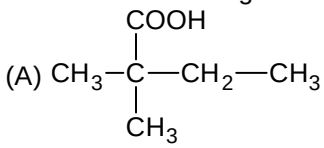
(A) 

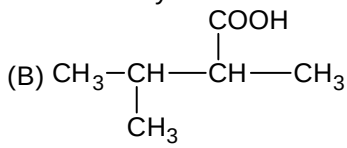
(B) 

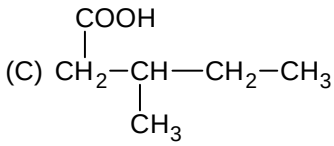
(C) 

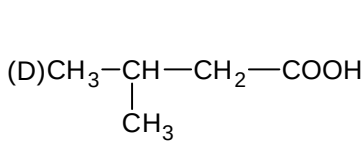
(D) 

28. Which of the following acid will give isopentane on decarboxylation with soda lime?

(A) 

(B) 

(C) 

(D) 

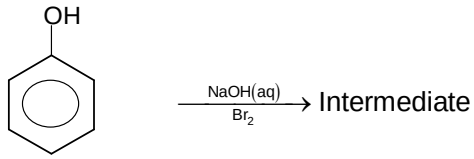
29. On treated with dilute H_2SO_4, a salt gives a gas which turns lime water milky. The salt can be

(A) $NaHCO_3$ (B) Na_2CO_3

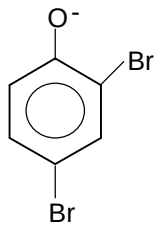
(C) Na_2SO_3 (D) $NaNO_2$

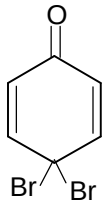
Space for Rough work

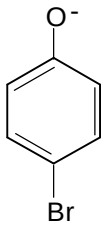
30. In the reaction

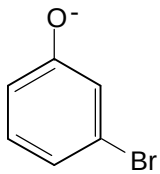


the intermediate are

(A) 

(B) 

(C) 

(D) 

SECTION – B

(Matching Type)

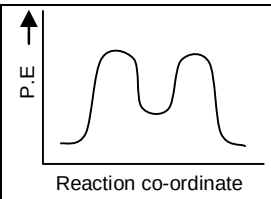
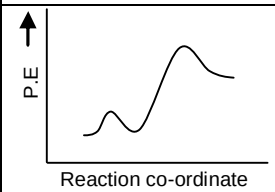
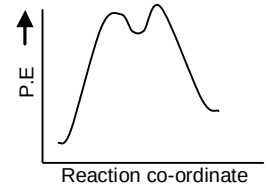
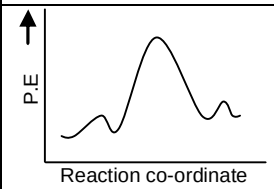
This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct.

31. Match the Column I and Column II:

Column I (Observation)		**Column II (Possible nuclear reaction)**	
(A)	α-emission	(p)	Z increase
(B)	β-emission	(q)	n/p increase
(C)	β^+-emission	(r)	n/p decrease
(D)	K-electron capture	(s)	X-ray emission
		(t)	Z decrease

Space for Rough work

32. Match the Column I and Column II:

Column I		**Column II**	
(A)		(p)	The intermediate are the final products
(B)		(q)	The reactants quickly form intermediate that slowly form products
(C)		(r)	The presence of intermediate is difficult to determine kinetically although the two step reaction is not ultra fast
(D)		(s)	The intermediate accumulate in measurable quantities and the reaction is easily resolved in two steps
		(t)	Mechansim like $E + S \underset{}{\overset{\text{fast}}{\rightleftharpoons}} Es$ $ES \rightleftharpoons EP$ $EP \rightleftharpoons E + P$

Space for Rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

33. In diamond, effective number of carbon atoms along any body diagonal of unit cell is n/4. What is the value of n.

34. One mole of SO_2Cl_2 on hydrolysis gives acids which can be neutralized by x moles of NaOH. The value of x is?

35. Ambidentate ligands out of
Py, en, CNS, CN, CNO, $C_2O_4^{--}$, NO_2 are

36. Brown coloured ring formed in nitrate test has formula $\left[Fe(H_2O)_5NO\right]SO_4$. Unpaired electron in Fe is

37. Taking one mole of $N_{2(g)}$ and excess of $H_2(g)$, under suitable conditions, formation of NH_3 is completed when $NH_3(g)$ has mole fraction of 0.5. The initial moles of H_2 are?

38. The mole fraction of solute in some solution is 1/n. If 50% of the solute molecules dissociates into two parts and remaining 50% get dimerized now mole fractions of solvent become 4/5. Find value of n.

39. Calculate the weight of MnO_2 required to react with HCl having specific gravity 1.2 gm/ml. 4% by mass necked to produce 1.8 L of Cl_2, at STP by reaction:

40. Total number of OH groups present on all boron atoms in colemanite $Ca_2B_6O_{11}.5H_2O$ is

Space for Rough work

Mathematics

PART – III

SECTION – A
One OR More Than One Choice Type

This section contains **10 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

41. If $f(x_0 - x_2), f(x_1)f(x_2) \& f(x_1 + x_2)$ are in A.P for all x_1, x_2 and $f(0) \neq 0$ then

(A) $f'(5) = f(-5)$ (B) $f(5) = f(-5)$

(C) $f'(5) - f'(-5) = 0$ (D) $f'(5) + f'(-5) = 0$

42. Consider a sequence $\{a_n\}$ with $a_1 = 2 \& a_n = \frac{a_{n-1}^2}{a_{n-2}}$ for all $n \geq 3$, terms of the sequence being distinct. If a_2 & a_5 are + ve integers and $a_5 \leq 162$, then the possible value (s) of a_5 can be

(A) 162 (B) 64

(C) 32 (D) 2

43. If $f_n(x) = e^{f_{n-1}(x)} \forall n \in N \& f_o(x) = x$, then $\frac{d}{dx} f_n(x)$ is equal to

(A) $f_n(x)\frac{d}{dx}(f_{n-1}(x))$ (B) $f_n(x)f_{n-1}(x)$

(C) $f_n(x)f_{n-1}(x).....f_2(x)f_1(x)$ (D) None of the above

44. Let ABC be a Δ with in centre at I. If P and Q are foot of the perpendicular from A to BI & CI resp. then which of the following results are correct?

(A) $\frac{AP}{BI} = \frac{\sin B/2 \cos C/2}{\sin A/2}$ (B) $\frac{AQ}{CI} = \frac{\sin C/2 \cos B/2}{\sin A/2}$

(C) $\frac{AP}{BI} = \frac{\sin C/2 \cos B/2}{\sin A/2}$ (D) $\frac{AP}{BI} + \frac{AQ}{CI} = \sqrt{3}$ if $\angle A = 60^o$

Space for Rough work

45. Let $\hat{u},\hat{v},\hat{w}$ be three unit vectors such that $\hat{u}+\hat{v}+\hat{w}=\hat{a}$, $\hat{a}.\hat{u}=\frac{3}{2}$, $\hat{a}.\hat{v}=\frac{7}{4}$ & $|\hat{a}|=2$, then

(A) $\hat{u}.\hat{v}=\frac{3}{4}$ (B) $\hat{u}.\hat{w}=0$

(C) $\hat{u}.\hat{v}=\frac{1}{4}$ (D) $\hat{u}.\hat{w}=-\frac{1}{4}$

46. Consider the integral $I_1=\int_1^e (1+x)(x+\ln x)^{100}\,dx$ $I_2=\int_{\sin^{-1}(1/e)}^{\pi/2} (1+e\sin x+\ln\sin x)^{101}\cos x\,dx$

If $I_1+\frac{e}{101}I_2=\frac{e(1+e)^{101}-k}{101}$ then $k \geq$

(A) 0 (B) 1

(C) 2 (D) – 1

47. The equation $k+x^3e^{x+3}=0$ has 2 distinct real roots & k is a prime natural number, then the possible value of k is/are

(A) 11 (B) 13

(C) 31 (D) 37

48. If the equation $\left(2^{\left(\frac{1}{\cos^{-1}x}\right)}\right)^{2\pi}-\left(a+\frac{1}{2}\right)\left(2^{\left(\frac{1}{\cos^{-1}x}\right)}\right)^{\pi}-a^2=0$ has only one real solution then subsets of values of 'a' are

(A) $(-3,1)$ (B) $(-\infty,-3]$

(C) $[1,\infty)$ (D) $[-3,\infty)$

Space for Rough work

49. Equation of the circles $|z-1-i|=1 \& |z-1+i|=1$ touches internally a circle of radius 2. The equation of the circle touching all the three circles can be

(A) $3z\bar{z}+z+\bar{z}-1=0$ (B) $3z\bar{z}-7\left(z+\bar{z}\right)+15=0$

(C) $z\bar{z}-z-\bar{z}-3=0$ (D) $3z\bar{z}+\left(z+\bar{z}\right)+1=0$

50. Let $a = \sin^{-1}(\sin 3)+\sin^{-1}(\sin 4)+\sin^{-1}(\sin 5)$, $f(x)=e^{x^2+|x|}$, domain of $f(x)$ be $[\alpha,\infty)$ & range of $f(x)$ be $[b,\infty)$ and $g(x)=\left(4\cos^4 x-2\cos 2x-\frac{1}{4}\cos 4x-x^7\right)^{1/7}$, domain & range of g(x) is set of real numbers. Which of the following are correct?

(A) $a=-2$ (B) $a+b=-1$

(C) $f\left(g\left(g(b)\right)\right)=e^2$ (D) Both f(x), g(x) are non invertible fns

SECTION – B

(Matching Type)

This section contains **2 multiple choice questions.** Each question has matching Column(s). The Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE THAN ONE** is/are correct.

51. Base of a pyramid is rectangular, three of its vertices of the base are $A(2,2,-1), B(3,1,2) \& C(1,1,1)$

Its vertex at the top is $P\left(4,\frac{-26}{3},\frac{-10}{3}\right)$ & fourth vertex of the base is D. Then

Column I		**Column II**	
(A)	Magnitude of position vector of D is	(p)	$\sqrt{120}$
(B)	Height of pyramid	(q)	$\sqrt{20}$
(C)	Volume of pyramid is	(r)	20
(D)	Area of base of pyramid is	(s)	60
		(t)	$\sqrt{60}$

Space for Rough work

52. Match the following

Column – I		Column – II	
(A)	$\frac{\prod_{k=1}^{100}\left(1-\tan^2\frac{2^k\pi}{2^{100}+1}\right)}{2^{100}}$ is equal to	(p)	+ 1
(B)	$\prod_{k=1}^{100}\left(1+2\cos\frac{2\pi.3^k}{3^{100}+1}\right)$ is equal to	(q)	- 1
(C)	Let n be a +ve integer and let x be a real number different from $2^{k+1}\left(\frac{\pi}{3}+l\ \pi\right)$, $k=1,2,.....n$, l an integer then $\left[\left(1+2\cos\frac{x}{2^{100}}\right)\prod_{k=1}^{100}\left(1-2\cos\frac{x}{2^k}\right)\right]-2\cos x$ is equal to	(r)	2
(D)	The value of $\sum_{n=1}^{\infty}\frac{1}{2^n}\tan\frac{2}{2^n}+\cot 2$ is equal to	(s)	3
		(t)	1/2

SECTION – C

(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

53. A tosses 2 fair coins & B tosses 3 fair cons after game is won by the person who throws greater number of heads. In case of a tie, the game is continued under identical rules until someone finally wins the game. The probability that A finally wins the game is $K/11$, then K is

54. The vertices of a triangle OBC are $0(0,0), B(-3,-1), C(-1,-3)$. Equation of line parallel to BC & intersecting the sides OB & OC whose perpendicular distance from the point (0, 0) is $\frac{1}{\sqrt{2}}$ $ax+by+2=0$ then the value of $\frac{a^4+b^4}{4}$ is

Space for Rough work

55. 8 Children are standing in a line outside a ticket counter at zoo. 4 of them have a 1 rupee coin each & the remaining four have a 2 rupee coin each. The entry ticket costs 1 rupee each. If all the arrangements of the 8 children are random, the probability that no child will have to wait for a change, if the cashier at the tickets window has no-change to start with is K. Then 15K is equal to

56. In $\Delta ABC, \angle ABC = 45^o$, point D is on segment BC such that $2|BD| = |CD|$ and $\angle DAB = 15^o$ & $\angle ACB$ is $(25\,k)^o$ then k =

57. Let n be a positive integer if $1 \le l < K \le n$, such that $\frac{\sin^2 nx}{\sin^2 x} = a_o + \sum_{1 \le l < k \le n} a_{l,k} \cos 2(k-l)$ for all real numbers x with x not an integer multiple of π, then the value of $a_{l,k}$ is

58. Given A = $\left\{ f : C[0,1] : \int_0^1 f(x)\,dx = 1 \right\}$ then minimum value of $\int_0^1 (1+x^2) f^2(x)\,dx, f \in A$ is $\frac{k}{\pi}$ then k is

59. Straight lines $3x - 4y = \frac{4}{\pi}\sin^{-1}(a^8+1) + 2\cos^{-1}(a^{12}+1) - \sec^{-1}(a^2+1), a \in R$ & $6x - 8y = 7$ are tangents to a circle then the length of arc of this circle which makes on angle of $\frac{40}{3}$ at its centre is

60. If $f(x)$ be a twice different able function from $R \to R$ such that $t^2 f(x) - 2tf'(x) + f''(x) = 0$ has two equal values of $t \forall x$ & $f(0) = 1, f'(0) = 2$ then $\lim_{x \to 0}\left(\frac{f(x)-1}{x} - \frac{t}{2}\right)$ is

Space for Rough work